
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ
IN THE NAME OF ALLAH

FINAL YEAR PROJECT | CDRB ADJUDICATION

# SDR-Based Doppler Radar

## using SDR for Velocity Measurement


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# Introduction

- Radar systems are fundamental tools for detecting object motion and measuring velocity.
- Traditional hardware radars are expensive, fixed-frequency, and difficult to reconfigure.
- Software Defined Radio (SDR) replaces dedicated hardware with flexible, software-controlled signal processing.
- The Analog Devices ADALM-PLUTO (PlutoSDR) is a low-cost, portable SDR platform operating from 325 MHz – 3.8 GHz.
- This project implements a complete Doppler radar system on SDR using GNU Radio for real-time velocity estimation.

# Problem Statement

*Conventional radar systems for velocity measurement are costly and hardware-dependent; a flexible, low-cost SDR-based Doppler radar is needed for accurate real-time velocity measurement without dedicated RF hardware.*

### Cost

Dedicated radar hardware is expensive

### Inflexibility

Fixed-frequency hardware cannot be reprogrammed

### Complexity

Separate TX/RX chains increase system complexity

# Proposed Solution

*Use SDR as a low-cost, software-reconfigurable platform to transmit a CW signal and capture the Doppler-shifted echo for real-time velocity computation via GNU Radio.*

### Low-Cost Hardware

- SDR
- No custom RF PCB needed
- USB-powered portable device

### Software-Defined Processing

- GNU Radio
- Real-time signal processing
- Reconfigurable frequency & gain

### Doppler Velocity Engine

- CW signal @ 2.482 GHz
- FFT-based frequency shift detection
- Velocity computed from fd = 2v·fc/c

# Doppler Effect – Principle

$$fd = 2 \cdot v \cdot fc / c$$

**fd** → Doppler frequency shift (Hz)

**v** → Target radial velocity (m/s)

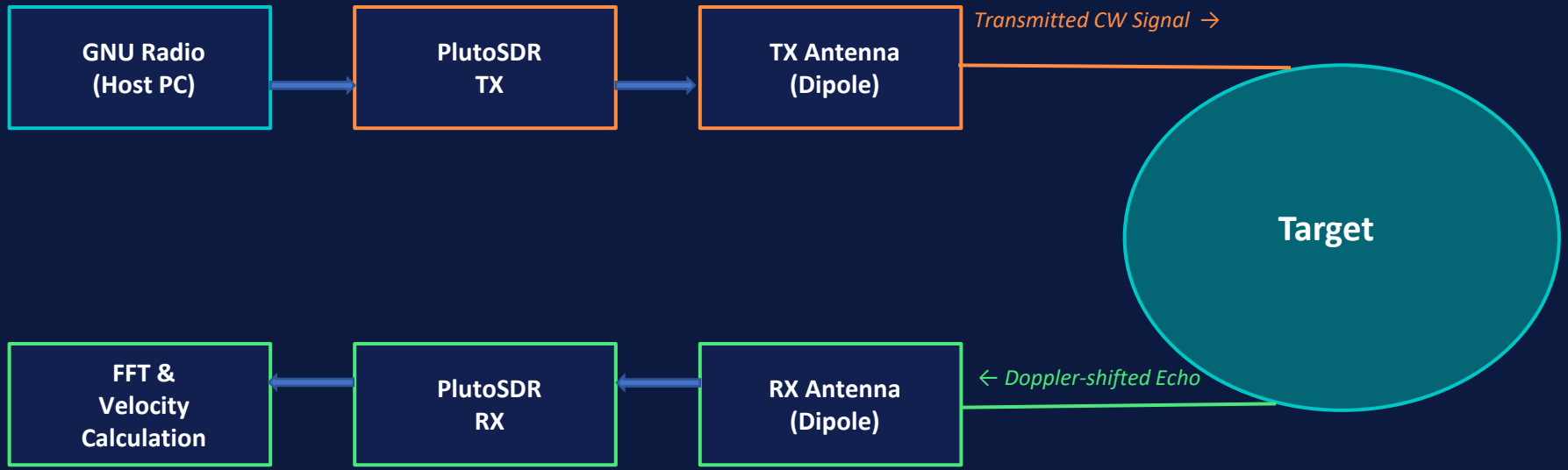
**fc** → Carrier frequency (Hz)

**c** → Speed of light = $3 \times 10^8$ m/s

## How It Works

- Radar transmits a continuous wave (CW) at fc = 2.482 GHz
- Moving target reflects the signal — reflected wave has a shifted frequency
- Frequency difference fd is proportional to velocity v
- Approaching targets → positive fd | Receding targets → negative fd

# System Block Diagram



# Signal Processing Flow

01

## CW Generation

GNU Radio generates a 2.482 GHz CW tone via the PlutoSDR TX

02

## Signal Transmission

Antenna radiates the signal; wave propagates to moving target

03

## Echo Reception

SDR RX captures reflected signal

04

## Mixing / Demodulation

Received signal is mixed to 0 Hz and decimated to 8 kS/s; Doppler shift appears as a low-frequency tone

05

## FFT Analysis

2048-point FFT, clutter map and peak detection

06

## Velocity Computation

$v = fd \times c / (2 \times fc)$ → shown live in GNU Radio and on the LED display

# SDR Platform – PLUTO (PlutoSDR)

| Parameter | Specification |
|---|---|
| RF Frequency Range | 325 MHz – 3.8 GHz |
| Bandwidth | Up to 20 MHz |
| ADC / DAC Resolution | 12-bit |
| Interface | USB 2.0 (High-Speed) |
| Transceiver IC | Analog Devices AD9363 |
| Carrier Freq (Used) | 2.482 GHz (ISM band) |
| TX Power | Up to +7 dBm |
| GNU Radio Support | gr-iio PlutoSDR blocks |

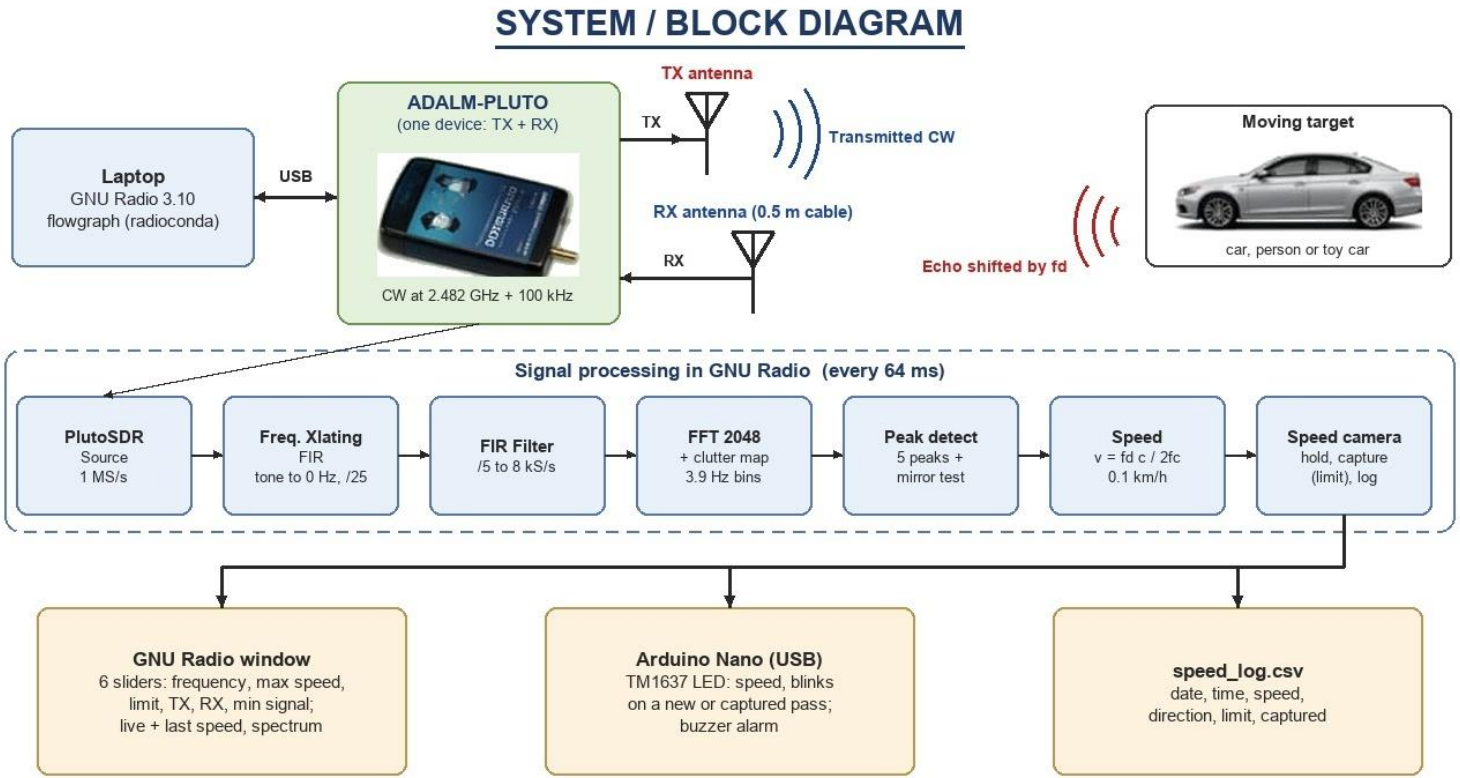
## Why PlutoSDR?

- Full-duplex transceiver — simultaneous TX and RX
- Wide tuning range — suitable for multiple radar bands
- Native GNU Radio support (gr-iio)
- Low cost — ideal for academic prototyping
- Compact form factor — portable deployment

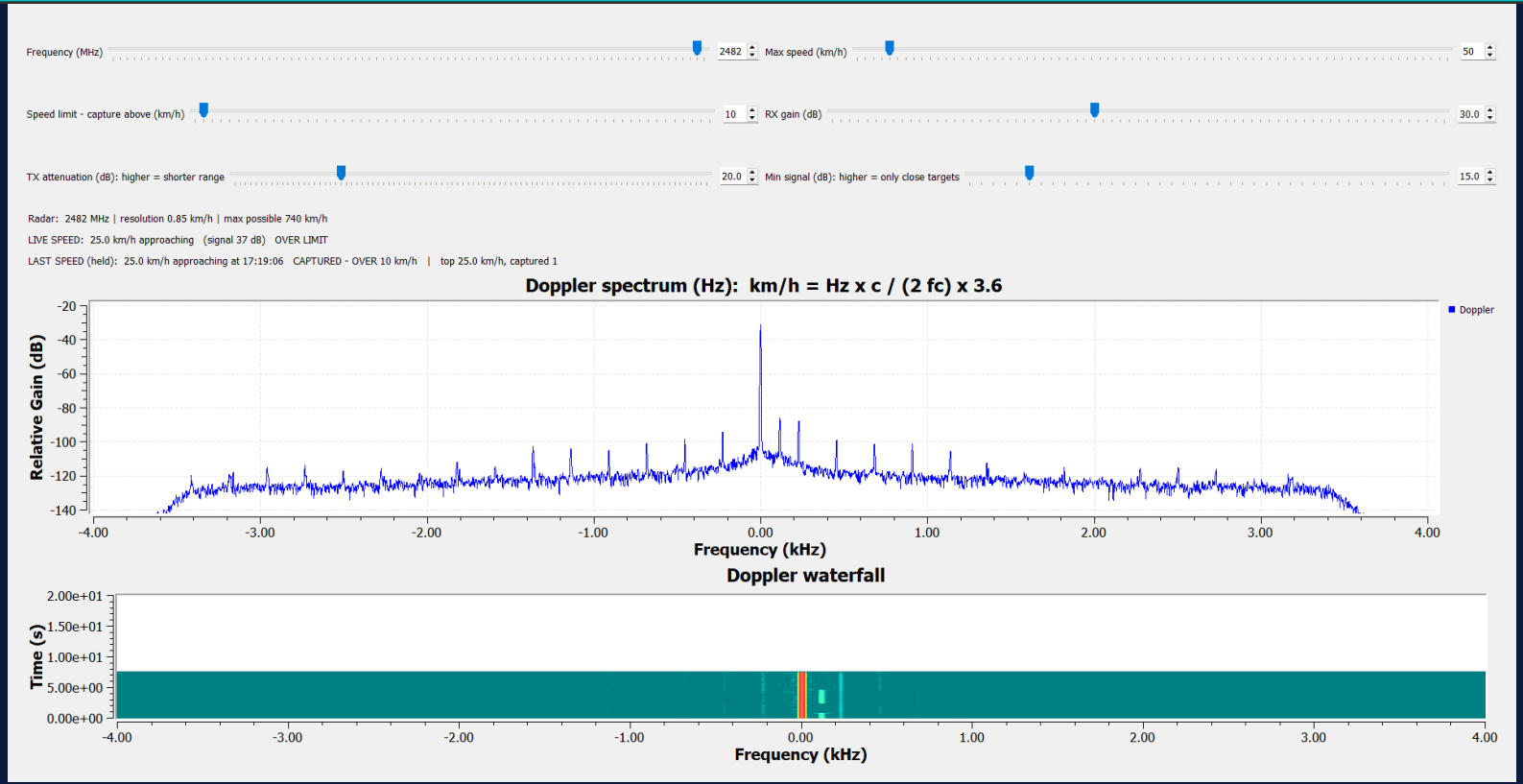
# System Specifications

| Parameter | Value | Note |
|---|---|---|
| Carrier Frequency | 2.482 GHz | ISM band; adjustable 2400–2483 MHz |
| Waveform Type | CW (Continuous Wave) | Simplest Doppler radar waveform |
| Sampling Rate | 1 MSPS | Decimated to 8 kS/s for Doppler |
| FFT Size | 2048 points | 0.26 s window, 3.9 Hz bins |
| Doppler Resolution | 0.85 km/h per bin | 0.1 km/h readout (peak interpolation) |
| Max Measurable Velocity | 50 km/h (default) | Adjustable up to 740 km/h (slider) |
| Antenna Type | Dipole (2.4 GHz) | ~2 dBi gain, omnidirectional |
| TX-RX Separation | ≥ 0.5 m | Reduce self-interference |
| Processing Platform | GNU Radio 3.10 | gr-iio PlutoSDR blocks + Python |

# Hardware Setup & System Diagram



# GNU Radio Implementation



# Test Results

| Test | Result | Details |
|---|---|---|
| Speed accuracy | Error below 0.05 km/h | Test targets 5–700 km/h injected through the Pluto |
| Direction | Always correct | Approaching (+) and receding (−) |
| Response time | 0.15 s live, 0.75 s held | Held speed on the LED after the target leaves |
| Empty room | No false detections | 90 s with nobody moving |
| Walking person | Detected, 3–13 km/h | Signal 24–47 dB above the noise |
| Toy car | Detected at close range | 3–17 km/h, driven straight at the antennas |
| Speed camera | Captured above the limit | Logged to CSV; LED keeps blinking |

# TENTATIVE BUDGET

| COMPONENTS |
| --- |
| Pluto SDR Device |
| Tx/Rx Antennas (2 units) |
| RF Components (cables, connectors) |
| Processing system (laptop) |
| Software tools (GNU Radio) |
| Miscellaneous (Arduino Nano, LED display, buzzer) |

# GANTT CHART - PROJECT TIMELINE



CONCLUSION

# A fully functional SDR-based Doppler radar — built and validated.

- ✅ Successfully implemented Doppler radar using PlutoSDR — no custom RF hardware required
- ✅ The system measures velocity from the Doppler shift in real time in GNU Radio and works as a speed camera with an LED display and adjustable frequency and speed limit
- ✅ The project demonstrate practical use of software defined radio in modern radar application with scope for future improvements such as range detection and higher accuracy


THANK
YOU